

Intelligent Document Extraction, Validation & API Platform

Invoices · Balance Sheets · Profit & Loss · Cash Flow Statements

OCR + layout-aware extraction · financial reconciliation · REST API · dashboard

The problem

The same numbers arrive in inconsistent, often scanned, layouts

1

Validate

Type, integrity and page limit —
before OCR

2

Read

Native PDF text, or rasterise and
OCR

3

Extract

Fields and every table row,
grounded

4

Reconcile

Recompute the financial
relationships

5

Serve

Store, expose via API and
dashboard

Four document types in scope

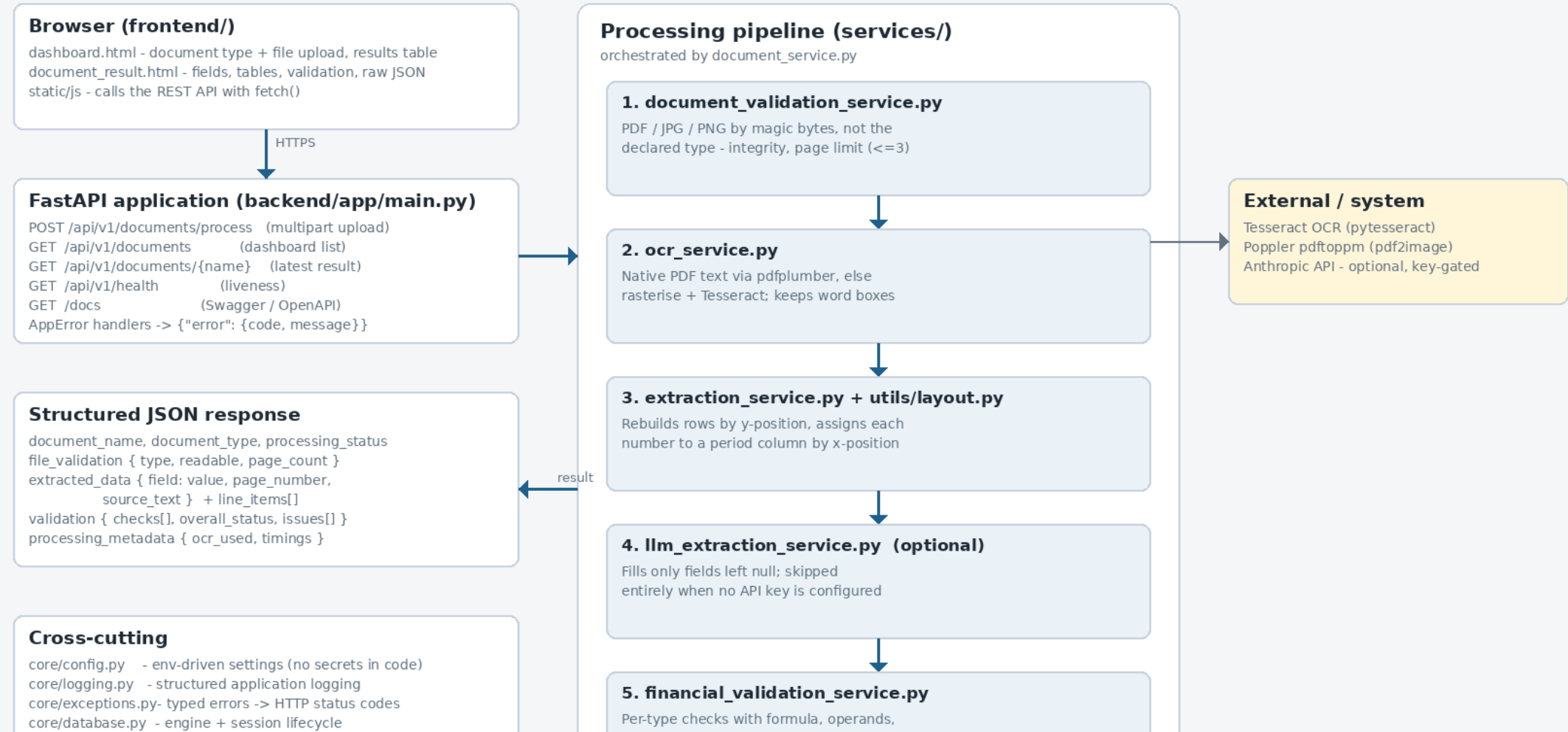
- Invoice — header fields, line items, tax, cash and change
- Balance sheet — capital & liabilities and assets, per period
- Profit & loss — income, expenditure, profit and appropriations
- Cash flow — operating, investing, financing, opening/closing cash

Architecture

One FastAPI application serves the API and the dashboard

Intelligent Document Extraction, Validation & API Platform

Request path for POST /api/v1/documents/process



Why line-by-line OCR text is not enough

The insight that drove the design

Reading OCR text as lines

Total 743, 732, 155

A row filled in for only one year gives no clue which year it belongs to — and OCR often emits labels and numbers as separate blocks entirely.

Reading the page as a table

Total · x≈1540 → column 2 → 31-Mar-16

Word boxes are kept from both PDF text and OCR. Rows are rebuilt by vertical position; each number is assigned to a period column by its right edge.

Measured, not assumed

- Benchmarked all 30 dataset statements; the first extractor was over-fitted to one year
- Compared page-segmentation modes and DPI — 300 DPI measured worse than 200
- Extraction and validation had drifted to two different sets of field lookups; unified them
- A second colour OCR pass gained one check for twice the latency, so it was left out

Financial validation

Each check reports formula, operands, calculated vs reported, variance, status

Invoice

$qty \times unit\ price \approx line\ amount \cdot \sum lines \approx subtotal \cdot subtotal + tax - discount \approx total \cdot cash - total \approx change$

Balance sheet

$Total\ Capital\ \&\ Liabilities \approx Total\ Assets \cdot \sum component\ rows \approx each\ reported\ total$

Profit & loss

$Interest + Other\ Income \approx Total\ Income \cdot Income - Expenditure \approx Net\ Profit \cdot - Minority \approx Attributable$

Cash flow

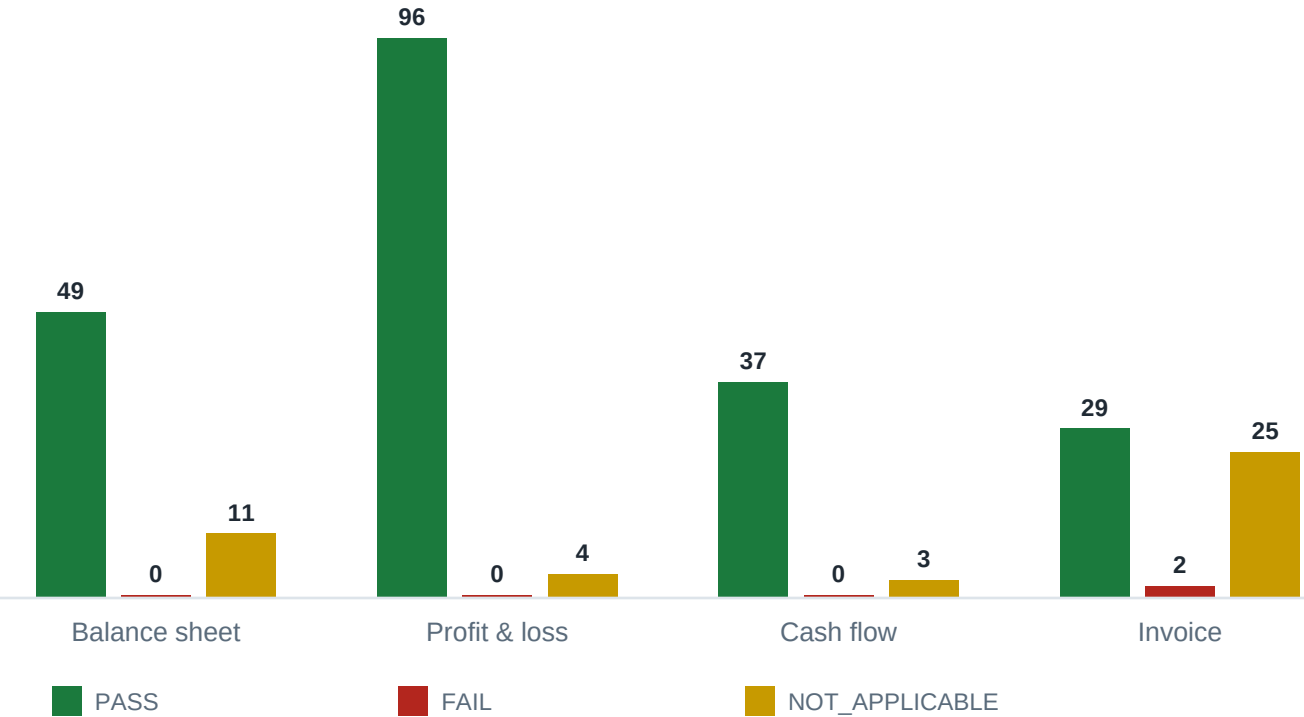
$Operating + Investing + Financing + FX \approx Net\ Increase \cdot Opening + Net\ Increase \approx Closing\ Cash$

Honest by design

- Missing operand → NOT_APPLICABLE, never a failure
- A partly-read section is not summed — an unread row would count as zero
- Tolerance: max(1.0, 1% of the larger value)
- Bracketed figures parse as negative
- Every check runs once per reporting period
- Missing values stay null — never guessed

Measured results

All 50 documents, 306 checks, with the Gemini vision pass



261

checks reconcile exactly

2

genuine discrepancies, verified by hand

43

not applicable — nothing to reconcile

Clean scans reconcile fully; the not-applicable results concentrate in the 2020–2022 scans where OCR cannot recover row labels.

API and dashboard

One deployment serves both; Swagger is generated from the code

POST	/api/v1/documents/process	upload and process
GET	/api/v1/documents	dashboard list
GET	/api/v1/documents/{name}	latest result
GET	/api/v1/health	health check
GET	/docs	Swagger / OpenAPI

Errors return one envelope

```
{ "error": { "code": "UNSUPPORTED_FILE_TYPE",
            "message": "Only PDF / JPG / PNG..." } }
```

415 · 413 · 400 · 404 · 422 · 500 — never a stack trace

Intelligent Document Extraction & Validation API

Back to dashboard

batch3-1495.jpg

Type: Invoice · Status: FAIL · Processed at: 9/12/2023, 5:25:07 AM · OCR used: yes

Extracted Fields

INVOICE NUMBER BEYOND DIGITAL IMAGING (p-1) 298.899 Not found	INVOICE DATE Oct. 13, 2023 (p-1) Not found	INVOICE TYPE ID Not found	CUSTOMER NAME Not found
SUBTOTAL Not found	TAX AMOUNT Not found	TAX RATE PERCENT Not found	TAX INCLUSIVE Not found
DISCOUNT Not found	TOTAL AMOUNT 5,257 (p-1)	TOTAL QUANTITY 27 (p-1)	CASH PAID Not found
CHANGE Not found			

Tables / Line Items

Line Items

Description	Qty	Unit price	Amount
3M 5.0/550 Dual Lock Fastener	9	367	3,303
3M 369 Trans Pads	9	1	9
3M 4949 V48 Acryl Foam	1	110	110
3M 2542 Electrical Padlock	6	4	24
3M 254 Perforate Flashback	2	17	34

Financial validation

Overall status: FAIL

Check	Formula	Calculated	Reported	Variance	Status
line_item_1_quantity_times_unit_price	quantity * unit_price = line amount	3,303	3,303	0	PASS
line_item_2_quantity_times_unit_price	quantity * unit_price = line amount	9	9	0	PASS
line_item_3_quantity_times_unit_price	quantity * unit_price = line amount	110	110	0	PASS
line_item_4_quantity_times_unit_price	quantity * unit_price = line amount	24	24	0	PASS
line_item_5_quantity_times_unit_price	quantity * unit_price = line amount	34	34	0	PASS
line_item_6_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_7_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_8_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_9_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_10_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_11_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_12_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_13_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_14_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_15_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_16_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_17_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_18_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_19_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_20_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_21_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_22_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_23_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_24_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_25_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_26_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_27_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_28_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_29_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_30_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_31_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_32_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_33_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_34_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_35_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_36_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_37_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_38_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_39_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_40_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_41_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_42_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_43_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_44_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_45_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_46_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_47_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_48_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_49_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_50_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_51_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_52_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_53_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_54_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_55_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_56_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_57_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_58_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_59_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_60_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_61_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_62_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_63_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_64_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_65_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_66_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_67_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_68_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_69_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_70_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_71_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_72_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_73_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_74_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_75_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_76_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_77_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_78_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_79_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_80_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_81_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_82_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_83_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_84_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_85_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_86_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_87_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_88_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_89_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_90_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_91_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_92_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_93_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_94_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_95_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_96_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_97_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_98_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_99_quantity_times_unit_price	quantity * unit_price = line amount				
line_item_100_quantity_times_unit_price	quantity * unit_price = line amount				

Raw JSON

Show/Hide

```
{
  "document_name": "batch3-1495.jpg",
  "document_type": "Invoice",
  "processing_status": "FAIL",
  "data_validation": {
    "title": "3M",
    "header": "3M",
    "line_items": "3M",
    "tax": "3M",
    "discount": "3M",
    "total": "3M",
    "change": "3M"
  },
  "extracted_data": {
    "invoice": {
      "number": "BEYOND DIGITAL IMAGING",
      "date": "2023-10-13",
      "source_text": "BEYOND DIGITAL IMAGING"
    },
    "customer": {
      "name": "3M Apple Creek Blvd",
      "address": "3M Apple Creek Blvd",
      "city": "Apple Creek",
      "state": "IL",
      "zip": "62506",
      "country": "USA"
    },
    "line_items": [
      {
        "description": "3M 5.0/550 Dual Lock Fastener",
        "quantity": 9,
        "unit_price": 367,
        "amount": 3303
      },
      {
        "description": "3M 369 Trans Pads",
        "quantity": 9,
        "unit_price": 1,
        "amount": 9
      },
      {
        "description": "3M 4949 V48 Acryl Foam",
        "quantity": 1,
        "unit_price": 110,
        "amount": 110
      },
      {
        "description": "3M 2542 Electrical Padlock",
        "quantity": 6,
        "unit_price": 4,
        "amount": 24
      },
      {
        "description": "3M 254 Perforate Flashback",
        "quantity": 2,
        "unit_price": 17,
        "amount": 34
      }
    ],
    "tax": {
      "rate": 0,
      "amount": 0,
      "inclusive": 0
    },
    "discount": {
      "amount": 0
    },
    "total": {
      "amount": 3480,
      "quantity": 27
    },
    "change": {
      "amount": 0
    }
  },
  "validation": {
    "line_items": "3M",
    "tax": "3M",
    "discount": "3M",
    "total": "3M",
    "change": "3M"
  }
}
```

Limitations and what I'd change

Known today

- OCR quality is the binding constraint — the 2020–2022 scans lose row labels entirely
- Statement parsing is tuned to this dataset's banking format
- Noisy receipts yield fewer line items
- Synchronous processing, ~2–6 s per scanned page
- No authentication — per case study scope

For production

- Queue the work and return a job id instead of holding the connection
- Commercial OCR, with the LLM pass standard rather than opt-in
- Golden-file regression suite over a labelled corpus
- Object storage for source documents; Alembic migrations
- Auth, rate limiting, metrics and alerting on the FAIL rate

Deterministic by default · every value traceable to its source row · nothing invented